

PROJECT PROPOSAL

Project Title: AI Based 5G KPI Anomaly Detection Dashboard for Telecom Networks

Business Problem

Modern telecom networks generate a massive amount of Key Performance Indicator (KPI) data from thousands of cells towers every few minutes. These KPIs, such as RSRP, SINR, Throughput, Latency, Packet Loss, and Connected Users, provide insights into the health and performance of the network.

Traditional network monitoring relies heavily on manual analysis and predefined threshold-based alerts. As telecom networks grow and complexity, these approaches become inefficient, leading to delayed fault detection, increased operational costs, and reduced Quality of Service (QoS). Undetected anomalies can result in poor user experience, service degradation, network congestion, and customer dissatisfaction.

To address these challenges, an intelligent AI-based anomaly detection system is required to automatically monitor network KPIs, identify abnormal behavior in real time, and provide actionable insights through an interactive dashboard.

Project Objective

The primary objective of this project is to develop an AI-powered web dashboard that continuously monitors telecom network KPIs and automatically detects anomalies using machine learning techniques.

The project aims to:

- Collect and preprocess telecom KPI data
- Perform feature engineering to improve model performance
- Train an Isolation Forest model for anomaly detection
- Evaluate the model using appropriate performance metrics
- Visualize network performance using an interactive Streamlit dashboard
- Display detected anomalies and KPI alerts in an intuitive manner
- Provide a scalable solution that assists network engineers in proactive network monitoring and fault detection

Project Scope

In Scope	Out Scope
• Telecom KPI data preprocessing • Data validation and cleaning	• Live integration with telecom base stations

• Feature engineering • Machine learning-based anomaly detection using Isolation Forest • Model evaluation • Hyperparameter tuning • Interactive Streamlit dashboard • KPI visualization using Plotly • Cell health monitoring • Network anomaly alerts • Project documentation and GitHub repository	• Automatic fault correction • Real-time deployment on production telecom infrastructure • Multi-user authentication and role management • Integration with OSS/BSS systems • Cloud deployment and distributed processing
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Expected Outcomes

At the completion of this project, the following outcomes are expected

- A clean and preprocessed telecom KPI dataset
- A trained Isolation Forest model capable of detecting anomalous network behavior
- Improved visibility into network performance through interactive visualizations
- Automatic identification of abnormal cell performance
- Faster detection of network degradation compared to manual monitoring
- A user-friendly Streamlit dashboard for network engineers
- Complete project documentation, evaluation reports, and source code hosted on GitHub

Assumptions

The project is based on the following assumptions

- The telecom KPI dataset accurately represents normal and abnormal network conditions
- KPI values are collected consistently over time
- The selected KPIs are sufficient to identify network anomalies
- Isolation Forest is suitable for unsupervised anomaly detection in telecom datasets
- Historical KPI data is available for model training
- The Streamlit dashboard will operate on a local machine using processed data.

Success Criteria

The project will be considered successful if it satisfies the following criteria:

- The preprocessing pipeline successfully cleans and prepares the dataset
- The anomaly detection model accurately identifies abnormal KPI patterns

- The model demonstrates reliable performance during evaluation
- The dashboard displays live KPI trends and detected anomalies correctly
- KPI alerts are generated for abnormal network conditions
- The application runs without errors and provides an intuitive user experience
- The project follows modular coding practices and is fully documented

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Isolation Forest
- Plotly
- Matplotlib
- Streamlit
- Joblib
- Git and GitHub
- VS Code

Expected Deliverables

- Project proposal
- Data preprocessing module
- Feature engineering module
- Trained isolation forest model
- Model evaluation report
- Hyperparameter comparison report
- Streamlit dashboard
- README Documentation
- GitHub Repository
- Final project report
- Project presentation
- Live demonstration